

Digital User Churn Dashboard

**Business Intelligence Analyst
Project Based Internship Program**

Presented by
Rifka Umdati



Rifka Umdati

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About

My capstone project for my Informatics degree involved developing an LSTM model for automated essay scoring. It was that project that first sparked my interest in taking data work more seriously. I've experienced many struggles since graduation when applying for internships and entry-level positions, including a time when I took a break to gather myself before starting over. During that period, I continued to develop my interest in data through individual projects, cleaning datasets and transforming them into insightful dashboards using Looker Studio, SQL, and Python. The opportunity to participate in this program came as a pleasant surprise, and I sincerely appreciate it and will make the most of it.

Experience & Education

Quality Assurance Intern (Juli-Agustus 2023)
Diskominfo Kab. Kuningan

- Cleaned and validated staple food price data for accurate reporting.
- Tested website functionality and UI/UX, documenting bugs and test cases.

Universitas Jenderal Soedirman (2021-2025)
Capstone Project:
Automated Essay Scoring system for assessing junior high school students' reading and writing skills using an LSTM deep learning model, deployed as a web application.

Case Study

PT Sejahtera Bersama needs an analysis of its sales performance based on transaction data collected over several periods. Using the four datasets provided: Customers, Products, Orders, and ProductCategory. This task involves building a set of visualizations to show:

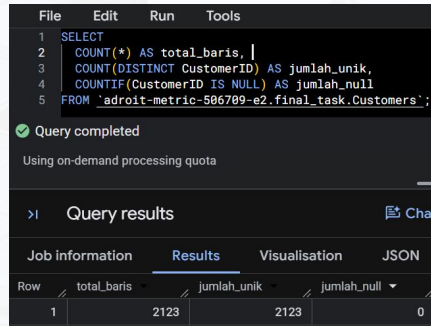
- Total overall sales
- Total sales by product category
- Total quantity sold by product category
- Total sales by city
- Total quantity sold by city
- Top 5 product categories by sales
- Top 5 product categories by quantity sold

From these visualizations, the goal is to identify patterns in the data and use them to come up with practical recommendations that could help maintain or increase sales going forward.

Question 1: Determine primary key for each tables

To determine whether the column included any duplicate values, a query utilizing COUNT(*), COUNT(DISTINCT column), and COUNTIF() was executed for each table. Additionally, a null check was performed to ensure that no values were missing. Both checks came back clean, confirming:

- Customers table: **CustomerID**
- Orders table: **OrderID**
- Products table: **ProdNumber**
- ProductCategory table: **CategoryID**

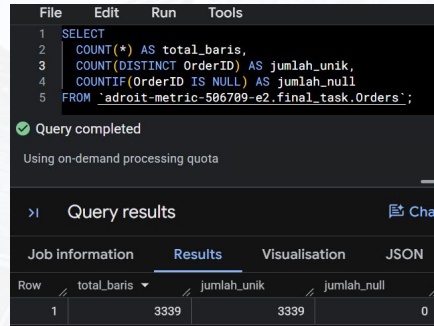


```
1 SELECT
2   COUNT(*) AS total_baris,
3   COUNT(DISTINCT CustomerID) AS jumlah_unik,
4   COUNTIF(CustomerID IS NULL) AS jumlah_null
5 FROM `adroit-metric-506709-e2.final_task.Customers`;
```

Query completed
Using on-demand processing quota

Query results

Job information	Results	Visualisation	JSON
Row	total_baris	jumlah_unik	jumlah_null
1	2123	2123	0

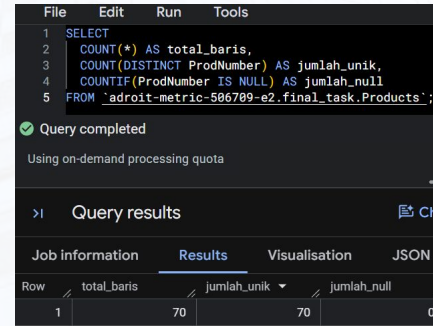


```
1 SELECT
2   COUNT(*) AS total_baris,
3   COUNT(DISTINCT OrderID) AS jumlah_unik,
4   COUNTIF(OrderID IS NULL) AS jumlah_null
5 FROM `adroit-metric-506709-e2.final_task.Orders`;
```

Query completed
Using on-demand processing quota

Query results

Job information	Results	Visualisation	JSON
Row	total_baris	jumlah_unik	jumlah_null
1	3339	3339	0

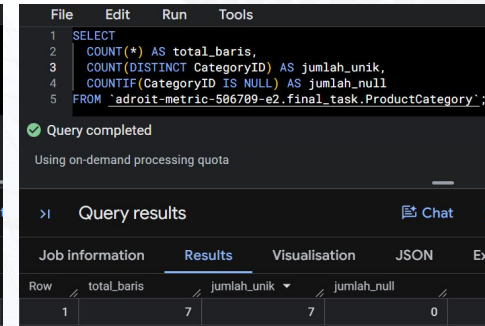


```
1 SELECT
2   COUNT(*) AS total_baris,
3   COUNT(DISTINCT ProdNumber) AS jumlah_unik,
4   COUNTIF(ProdNumber IS NULL) AS jumlah_null
5 FROM `adroit-metric-506709-e2.final_task.Products`;
```

Query completed
Using on-demand processing quota

Query results

Job information	Results	Visualisation	JSON
Row	total_baris	jumlah_unik	jumlah_null
1	70	70	0



```
1 SELECT
2   COUNT(*) AS total_baris,
3   COUNT(DISTINCT CategoryID) AS jumlah_unik,
4   COUNTIF(CategoryID IS NULL) AS jumlah_null
5 FROM `adroit-metric-506709-e2.final_task.ProductCategory`;
```

Query completed
Using on-demand processing quota

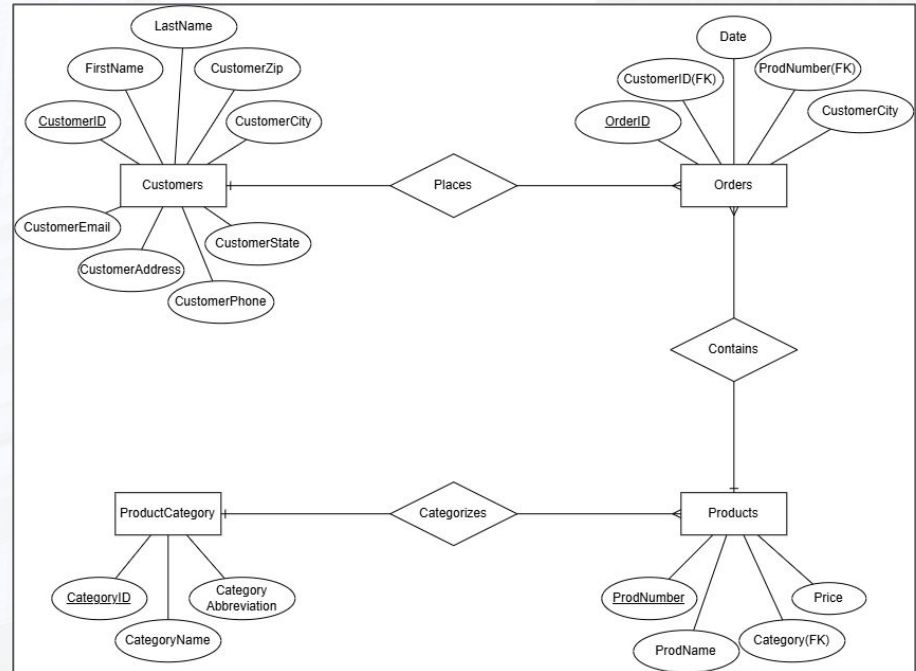
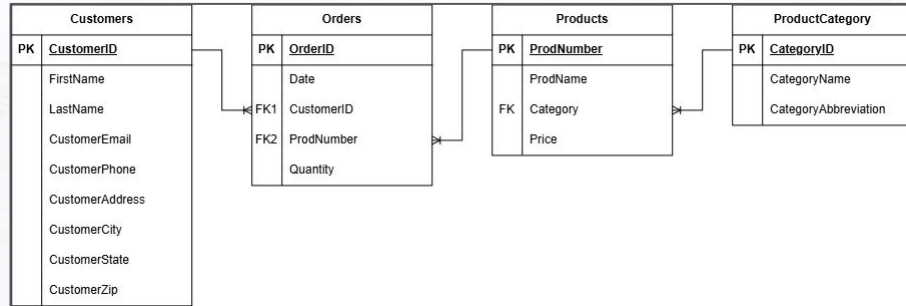
Query results

Job information	Results	Visualisation	JSON	Ex
Row	total_baris	jumlah_unik	jumlah_null	
1	7	7	0	

Question 2: Identify each table's relationships

There are 3 relationships between the 4 tables (All three are one-to-many relationships.):

- Customers to Orders: one customer can have many orders (linked via CustomerID)
- Products to Orders: one product can appear in many orders (linked via ProdNumber)
- ProductCategory to Products: one category can have many products (linked via CategoryID/Category)



Question 3: Create a Master Table

Before creating the dashboard in Looker Studio, the four tables were combined into a single master table using BigQuery, based on the relationships identified in Question 2. During the process, a formatting issue was identified in the price data, where commas were used instead of decimal points, causing BigQuery to interpret the values incorrectly. The issue was corrected before the final query was executed. The resulting master table was then sorted by transaction date, from the earliest to the latest, and exported as a CSV file to be used as the data source for the Looker Studio dashboard.

```
1 SELECT
2   o.Date AS order_date,
3   pc.CategoryName AS category_name,
4   p.ProdName AS product_name,
5   p.Price AS product_price,
6   o.Quantity AS order_qty,
7   o.Quantity * p.Price AS total_sales,
8   c.CustomerEmail AS cust_email,
9   c.CustomerCity AS cust_city
10  FROM `adroit-metric-506709-e2.final_task.Orders` o
11  JOIN `adroit-metric-506709-e2.final_task.Customers` c ON o.CustomerID = c.CustomerID
12  JOIN `adroit-metric-506709-e2.final_task.Products` p ON o.ProdNumber = p.ProdNumber
13  JOIN `adroit-metric-506709-e2.final_task.ProductCategory` pc ON p.Category = pc.CategoryID
14  ORDER BY o.Date ASC, o.OrderID ASC
```

Job information		Results	Visualisation	JSON	Execution details		Execution graph	
Row	order_date	category_name	product_name	product_price	order_qty	total_sales	cust_email	cust_city
1	2020-01-01	eBooks	Polar Robots	2399	2	4798	fvaslerqt@comsenz.com#mailto:...	Jackson
2	2020-01-01	Robots	RWW-75 Robot	88300	3	264900	tmckernot@tinyurl.com#mailto:...	Katy
3	2020-01-01	Training Videos	Drone Video Techniques	3799	6	22794	gstiggersdd@eventbrite.com#mailto:gstiggersdd@eventbrite.com#	Saint Petersburg
4	2020-01-01	Drone Kits	BYOD-220	6900	1	6900	edew@nba.com#mailto:edew@...	Honolulu
5	2020-01-01	eBooks	SCARA Robots	1950	5	9750	llespercx@com.com#mailto:lle...	Des Moines
6	2020-01-01	eBooks	Spherical Robots	1675	5	8375	lfromonte9@de.vu#mailto:fro...	Birmingham
7	2020-01-02	Robot Kits	BYOR-2640S	18900	2	37800	aguilongo@behance.net#mailto:...	Houston

Question 4: Create a Dashboard

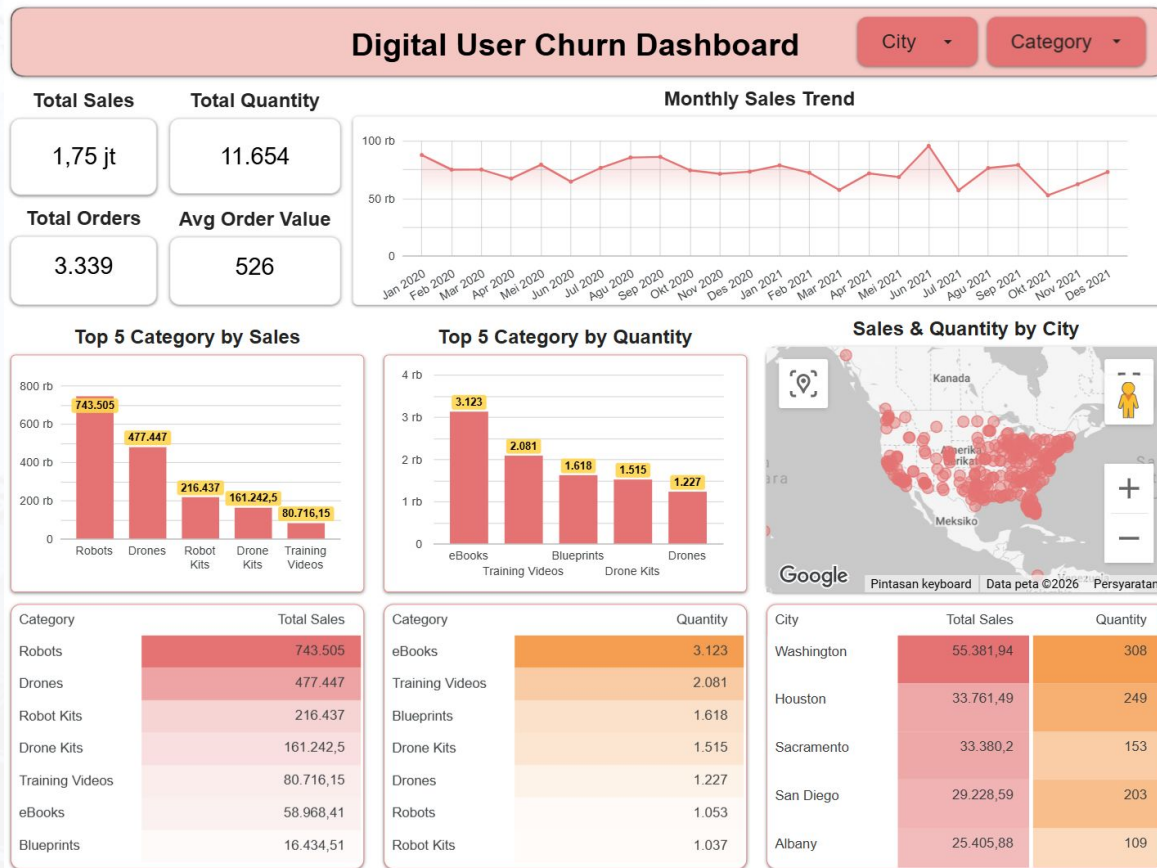
The "Digital User Churn Dashboard" created in Looker Studio, has all the necessary components: Total Sales and Total Quantity, sales and quantity by category, sales and quantity by city (Sales & Quantity by City), and Top 5 Category by Sales and Top 5 Category by Quantity. A few extra elements were also added: Monthly Sales Trend, Avg Order Value, Total Orders, and filters for city and category so the data can be explored further.

Main Takeaways:

- Total sales reached \$1,754,750.57, with a total of 11,654 units sold overall.
- The most popular categories in terms of sales are Robots, Drones, Robot Kits, Drone Kits, and Training Videos. But when it comes to amount sold, the most popular items are eBooks, Training Videos, Blueprints, Drone Kits, and Drones.
- Washington stands out as the top-performing city, with the highest sales and order quantity among all cities.
- Looking at the monthly sales trend, the business has shown some instability since 2020, with noticeable fluctuations from month to month, especially throughout 2021.

Question 4: Create a Dashboard

This Dashboard can be accessed through this [link](#).



Question 5: Business Recommendation

Based on the insights obtained from the dashboard, there are several recommendations that can be proposed to improve the company's business performance:

- **Bundle expensive products with cheaper ones that sell often**
Robots and Drones bring in the most money, but people don't buy them very often. Meanwhile, eBooks and Training Videos are bought a lot, just at a much lower price. Putting these together as a bundle might get customers who were only planning to buy the cheap item to also buy the more expensive one.
- **Put more marketing focus on Washington, Houston, and Sacramento**
These three cities already have the most sales and orders, so there are probably already a lot of customers there who are used to buying from this business. It seems more worth it to focus on these cities first with promotions or ads, instead of spending the same amount everywhere.
- **Make sure stock doesn't run out for the most-ordered products**
eBooks, Training Videos, Blueprints, Drone Kits, and Drones are ordered the most, even if they're not the highest in sales value. If these run out of stock, it could actually hurt sales more than expected, since customers are clearly buying them often.
- **Check what caused the ups and downs in 2021**
Sales moved up and down more in 2021 compared to 2020. It's not clear from this data alone why that happened – it could be something seasonal, a certain category doing worse, or something else going on that month. This would need to be looked into more before deciding what to do about it.
- **Tracking whether customers come back to buy again**
Right now, the data only shows individual transactions, not whether the same person bought more than once. Knowing this would help tell if sales are growing because of new customers or because old customers keep coming back, which matters for figuring out what to focus on next.

Github Project Link Here

[ririfka08/FinalTask_BI-Analyst_BankMuamalat_PBI-Rakamin](#)

File & Video Presentation Here

<https://drive.google.com/drive/folders/1BhrVqrP9Y1llyouL6TYK0I7iNF9tGSTbJ?usp=sharing>

Thank You



Rakamin
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